

Development Tools and Environment

Resilio — Mobile Mental Health Platform for Nepal | Milan Shrestha | ID: 23057135 | 03_Construction/Development_Tools_and_Environment.md

All tools, software, services, and environment configurations used during the development of Resilio. Organised by category.

INTEGRATED DEVELOPMENT ENVIRONMENTS (IDEs)

Android Studio

Primary IDE for Flutter development.

Attribute	Detail
Version	Android Studio Hedgehog (2023.1.1) or later
Purpose	Flutter/Dart mobile frontend development
Key plugins installed	Flutter plugin, Dart plugin, Dart DevTools
Download	https://developer.android.com/studio

Why Android Studio over VS Code for Flutter:

Android Studio's Flutter plugin provides a superior widget inspector, built-in Android emulator management, and Gradle build system integration — all essential for Android-first Flutter development.

Visual Studio Code

Secondary editor for Node.js backend development.

Attribute	Detail
Version	Latest stable
Purpose	Node.js / Express backend, SQL scripts, Markdown

Attribute	Detail
Key extensions installed	ESLint, Prettier, REST Client, GitLens, Thunder Client, Markdown Preview Enhanced
Download	https://code.visualstudio.com

FLUTTER / DART TOOLCHAIN

Tool	Version	Purpose
Flutter SDK	3.x (stable)	Cross-platform mobile framework
Dart SDK	Bundled with Flutter 3.x	Programming language
<code>`flutter pub get`</code>	—	Install Dart/Flutter packages
<code>`build_runner`</code>	Latest	Code generation: Freezed, json_serializable, injectable
Dart DevTools	Bundled	Widget inspector, performance profiler, memory analysis
<code>`flutter analyze`</code>	Bundled	Static analysis and linting
<code>`flutter test`</code>	Bundled	Unit and widget test runner

Common build commands used during development:

```
# Install all packages
flutter pub get

# Run code generation (Freezed, json_serializable, injectable)
flutter pub run build_runner build --delete-conflicting-outputs

# Run the app on connected device/emulator
flutter run

# Run all tests
flutter test

# Build release APK
flutter build apk --release

# Analyse for warnings and errors
flutter analyze
```

NODE.JS BACKEND TOOLCHAIN

Tool	Version	Purpose
Node.js	20.x LTS	JavaScript runtime
npm	Bundled with Node.js	Package manager
nodemon	Latest	Auto-restart server on file changes during development
eslint	Latest	JavaScript linting
prettier	Latest	Code formatting

Common commands used:

```
# Install dependencies
npm install

# Start development server with auto-restart
npm run dev # uses nodemon

# Start production server
npm start

# Lint check
npm run lint
```

VERSION CONTROL

Git + GitHub

Attribute	Detail
Tool	Git 2.x
Remote	GitHub (private repository)
Branching strategy	Feature branches → `main`; branch per major feature/iteration
Commit convention	`feat:`, `fix:`, `enhc:`, `refactor:` prefixes

Attribute	Detail
Download	https://git-scm.com

Branching structure used:

```
main                ← stable, submitted code
■■■ enhc/customer-enhanced
■■■ feat/webrtc-video-call
■■■ feat/esewa-payment
■■■ feat/mood-tracking
■■■ feat/admin-dashboard
■■■ feat/offline-sync
```

API TESTING

Postman

Attribute	Detail
Version	Latest
Purpose	Manual testing of all REST API endpoints during development
Collections created	Auth, Appointments, Mood, Questionnaires, Content, Admin, Payment
Environment variables	`BASE_URL`, `AUTH_TOKEN`, `TEST_USER_ID`, `TEST_THERAPIST_ID`
Download	https://www.postman.com

How Postman was used:

- Testing each endpoint immediately after implementation before writing unit tests
- Simulating eSewa payment callbacks with tampered amounts to verify rejection
- Testing RLS by swapping auth tokens between users
- Documenting request/response formats (see 07_API_Reference.md)

Thunder Client (VS Code Extension)

Used as a lightweight in-editor alternative to Postman for quick endpoint checks without switching windows.

DATABASE MANAGEMENT

Supabase Dashboard

Attribute	Detail
URL	https://supabase.com/dashboard
Purpose	Table management, SQL editor, RLS policy editor, query logs

Key tasks done in Supabase Dashboard:

- Creating all 20 database tables with constraints
- Writing and testing RLS policies using the Policy Editor
- Running SQL migration scripts
- Monitoring query performance
- Inspecting foreign key relationships
- Checking real-time subscription events

DBeaver

Attribute	Detail
Version	Community Edition, latest
Purpose	Desktop GUI for PostgreSQL — running complex SQL queries, viewing table relationships visually, data export
Download	https://dbeaver.io

Why DBeaver alongside Supabase Dashboard:

The Supabase dashboard SQL editor is useful but limited for complex multi-table queries and schema visualisation. DBeaver provides a full ER diagram view and a more capable SQL editor for development-time database work.

DESIGN TOOLS

Figma

Attribute	Detail
Purpose	High-fidelity UI mockups, component library, screen flow prototyping
Account type	Free tier (sufficient for project scope)
Files created	Resilio — Customer Module; Resilio — Therapist Module; Resilio — Admin Module; Design System
Link	https://figma.com

Key design decisions made in Figma:

- Established the colour palette: teal-green primary (#2DB5A3), off-white backgrounds, warm accent
- Created reusable component library: buttons, cards, input fields, bottom sheets
- Prototyped the full onboarding flow for client review in Week 7
- Exported all assets (icons, illustrations) as SVG/PNG for Flutter integration

draw.io / diagrams.net

Attribute	Detail
Purpose	All UML diagrams, ERD, DFD, system architecture diagram, WBS
Export format	PNG for report insertion; XML source for editability
Link	https://app.diagrams.net

Diagrams produced:

- System architecture (Flutter → Node.js → Supabase → External Services)
- Entity-Relationship Diagram (all 20 tables)
- Use case diagrams for Customer, Therapist, Admin
- Activity diagrams (booking flow, WebRTC flow, content publication)
- Level 0 and Level 1 DFDs
- Work Breakdown Structure

Mermaid Live Editor

Attribute	Detail
Purpose	Sequence diagrams embedded in report as Mermaid code
Link	https://mermaid.live

All sequence diagram source code is in FYP Report Writing/UML Diagrams (Mermaid Code).md.

Lucidchart

Attribute	Detail
Purpose	Gantt chart (initial and final), project timeline visualisation
Link	https://lucidchart.com

TESTING ENVIRONMENT

Android Emulator (Android Studio)

Configuration	Detail
Device profile	Pixel 6 (API 34, Android 14)
Purpose	Primary development and functional testing
Limitation	Cannot fully test push notifications (FCM); WebRTC sometimes behaves differently than real device

Physical Android Device

Device	OS Version	Purpose
Personal Android phone	Android 13	Testing push notifications (FCM requires real device), WebRTC video/audio, eSewa payment app integration, offline mode (airplane mode testing)

Tests that required a physical device:

- FCM push notification receipt (TC-U026, TC-S011)
- WebRTC video and audio quality assessment (TC-S004)
- eSewa SDK payment flow (TC-U014, TC-S003)
- Offline mood logging with airplane mode (TC-UAT023)

PROJECT MANAGEMENT

Notion

Attribute	Detail
Purpose	Project planning, sprint backlog, weekly task tracking
Link	https://notion.so

What was tracked in Notion:

- Sprint backlog for each Construction iteration (Iterations 1–7)
- Feature status: To Do → In Progress → Done
- Bug log with priority ratings
- Research notes and source organisation
- Weekly supervisor meeting preparation notes

COMMUNICATION AND COLLABORATION

Tool	Purpose
WhatsApp	Communication with external clients (Ms. Karki, Dr. Singh)
Zoom	Video interview with Dr. Dibyandra Singh (Week 2)
Google Forms	Pre-survey and post-survey distribution
Google Drive	Document sharing with supervisor
Email	Formal correspondence; supervisor feedback

ENVIRONMENT CONFIGURATION

Flutter Environment

```
# pubspec.yaml (key dependencies)
environment:
  sdk: '>=3.0.0 <4.0.0'
  flutter: '>=3.10.0'
```

Node.js Environment Variables (.env)

```
# Server
PORT=3000
NODE_ENV=development

# Supabase
SUPABASE_URL=https://[project-ref].supabase.co
SUPABASE_SERVICE_KEY=[service-role-key]

# SuperTokens
SUPERTOKENS_CONNECTION_URI=https://[instance].aws.supertokens.io
SUPERTOKENS_API_KEY=[api-key]

# Firebase
FIREBASE_SERVICE_ACCOUNT_PATH=./firebase-service-account.json

# Cloudinary
CLOUDINARY_CLOUD_NAME=[cloud-name]
CLOUDINARY_API_KEY=[key]
CLOUDINARY_API_SECRET=[secret]

# eSewa
ESEWA_SECRET_KEY=[secret]
ESEWA_CLIENT_ID=[client-id]
```

Security note: The .env file is in .gitignore and was never committed to version control. All secrets are stored in environment variables or the deployment platform's secret manager.

HARDWARE USED

Hardware	Specification	Use
Development laptop	MacOS 14, Apple M-series chip	Primary development machine

Hardware	Specification	Use
Personal Android phone	Android 13	Physical device testing
External display	27" monitor	Extended workspace for running emulator + IDE simultaneously